

Week3 Practical

Task 1) Document Modeling — create a book_metadata collection with embedded patterns (reviews list inside the book document) and referenced patterns (linking to member_id in SQL). 2) Advanced Querying — use comparison operators (\$gt, \$in) to find books rated above 4, and a regex query to find books with "Digital" formats. 3) Aggregation Pipeline — match books published after 2020, \$unwind the reviews array, then \$group by title and compute the average rating. 4) Indexing for Speed — create a text index on the review comments field and use .explain() to check whether the query uses the index.

1) Document Modeling

The screenshot shows the MongoDB Atlas interface for the 'book_metadata' collection. The 'Documents' tab is active, showing two documents. The first document is for 'The Silent Algorithm' by R. Menon, published in 2022, with an average rating of 4.6. The second document is for 'Ocean of Code' by A. Fernandes, published in 2021, with an average rating of 4.2. Both documents have embedded 'reviews' arrays and 'formats' arrays.

Student > BookFlow > book_metadata Open MongoDB shell

Documents 4 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) Explain Reset Find Options

ADD DATA BULK EXPORT DATA EXPORT CODE 25 1 - 4 of 4

```
{
  "_id": ObjectId('6a8d08239998e4beab722eca'),
  "title": "The Silent Algorithm",
  "author": "R. Menon",
  "genre": "Technology",
  "publishedYear": 2022,
  "formats": Array (3),
  "specs": Object (5),
  "avgRating": 4.6,
  "reviews": Array (2)
}
```

```
{
  "_id": ObjectId('6a8d08309998e4beab722ecc'),
  "title": "Ocean of Code",
  "author": "A. Fernandes",
  "genre": "Technology",
  "publishedYear": 2021,
  "formats": Array (2),
  "specs": Object (3),
  "avgRating": 4.2,
  "reviews": Array (2)
}
```

```
book_metadata | mongosh: Student +
>_MONGOSH
> use("BookFlow")
< switched to db BookFlow
> db["book_metadata"].find()
< {
  _id: ObjectId('6a8d08239998e4beab722eca'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
```

Task2: Advanced Querying

```
{
  "avgRating": {
    "$gt": 4
  }
}
```

[Generate query](#) [Explain](#) [Reset](#) [Find](#) [Options](#)

ADD DATA

BULK

EXPORT DATA

EXPORT CODE

25

1 - 3 of 3

```
_id: ObjectId('6a8d08239998e4beab722eca')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Array (2)
```

```
_id: ObjectId('6a8d08309998e4beab722ecc')
title: "Ocean of Code"
author: "A. Fernandes"
genre: "Technology"
publishedYear: 2021
formats: Array (2)
specs: Object (3)
avgRating: 4.2
reviews: Array (2)
```

Documents 4

Aggregations

Schema

Indexes 1

Validation

```
{
  "formats": {
    "$in": [
      "Digital",
      "Audiobook"
    ]
  }
}
```

Generate query ↗

Explain

Reset

Find

Options ▶

ADD DATA

BULK

EXPORT DATA

EXPORT CODE

251 - 3 of 3

```
_id: ObjectId('6a8d08239998e4beab722eca')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Array (2)
```

```
_id: ObjectId('6a8d08309998e4beab722ecc')
title: "Ocean of Code"
author: "A. Fernandes"
genre: "Technology"
publishedYear: 2021
formats: Array (2)
```

Documents 4

Aggregations

Schema

Indexes 1

Validation

```
{
  "avgRating": {
    "$gt": 4
  },
  "formats": {
    "$in": [
      "Digital"
    ]
  }
}
```

Generate query ↗

Explain

Reset

Find

Options ▶

ADD DATA

BULK

EXPORT DATA

EXPORT CODE

251 - 3 of 3

```
_id: ObjectId('6a8d08239998e4beab722eca')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Array (2)
```

```
_id: ObjectId('6a8d08309998e4beab722ecc')
title: "Ocean of Code"
author: "A. Fernandes"
genre: "Technology"
publishedYear: 2021
```


Task3: Aggregation Pipeline

▼

\$match

\$unwind

\$group

\$sort

\$project

Generate aggregation Explain

Untitled – modified SAVE CREATE NEW EXPORT DATA EXPORT CODE PREVIEW STAGES

▼ Stage 5 \$project

```
1 {
2   "_id": 0,
3   "title": "$_id",
4   "author": 1,
5   "avgReviewRating": {
6     "$round": [
7       "$avgReviewRating",
8       2
9     ]
10  },
11  "totalReviews": 1
12 }
```

Output preview after \$project stage (Sample of 3 documents)

```
author : "R. Menon"
totalReviews : 2
title : "Quantum Horizons"
avgReviewRating : 5
```

▼

\$match

\$unwind

\$group

\$sort

\$project

Edit

Explain Run Options

ALL RESULTS EXPORT DATA EXPORT CODE Showing 1 – 3 count results < > ▼ ☰

```
author : "R. Menon"
totalReviews : 2
title : "Quantum Horizons"
avgReviewRating : 5
```

```
author : "R. Menon"
totalReviews : 2
title : "The Silent Algorithm"
avgReviewRating : 4.5
```

```
author : "A. Fernandes"
totalReviews : 2
title : "Ocean of Code"
avgReviewRating : 4.5
```

Task4: Indexing for Speed

Documents 4AggregationsSchemaIndexes 2Validation

Create IndexRefreshVIEWINGINDEXESSearch Indexes

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	20.5 kB	7 (since Tue Aug 25 2026)	UNIQUE	READY
> reviews.comment_text	TEXT	20.5 kB	0 (since Tue Aug 25 2026)		READY

Documents 4AggregationsSchemaIndexes 2Validation

Generate query + ExplainResetFindOption

ADD DATABULKEXPORTE DATAEXPORT CODE251 - 3 of 3

```
{
  "$text": {
    "$search": "Digital"
  }
}
```

```
_id: ObjectId('6a8d08239998e4beab722eca')
title: "The Silent Algorithm"
author: "R. Menon"
genre: "Technology"
publishedYear: 2022
formats: Array (3)
specs: Object (5)
avgRating: 4.6
reviews: Array (2)
```

```
_id: ObjectId('6a8d08519998e4beab722ed0')
title: "Quantum Horizons"
author: "R. Menon"
genre: "Science"
publishedYear: 2023
formats: Array (2)
specs: Object (4)
avgRating: 4.9
reviews: Array (2)
```

Documents 4AggregationsSchemaIndexes 3Validation

Create IndexRefreshVIEWINGINDEXESSearch Indexes

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	20.5 kB	7 (since Tue Aug 25 2026)	UNIQUE	READY
> reviews.comment_text	TEXT	20.5 kB	6 (since Tue Aug 25 2026)		READY
> formats_1	REGULAR	20.5 kB	0 (since Tue Aug 25 2026)		READY

Documents 4 Aggregations Schema **Indexes 4** Validation

Create Index Refresh

VIEWING INDEXES SEARCH

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR ⓘ	20.5 kB	7 (since Tue Aug 25 2026)	UNIQUE ⓘ	READY
> reviews.comment_text	TEXT ⓘ	20.5 kB	6 (since Tue Aug 25 2026)		READY
> formats_1	REGULAR ⓘ	20.5 kB	0 (since Tue Aug 25 2026)		READY
> avgRating_-1	REGULAR ⓘ	20.5 kB	0 (since Tue Aug 25 2026)		READY

Documents 4 Aggregations Schema Indexes 4 Validation

⌵

{
 "avgRating": {
 "\$gt": 4
 }
}

Generate query ⚡ Explain Reset Find

➕ ADD DATA ▾ 📄 BULK ▾ 📄 EXPORT DATA ▾ </> EXPORT CODE

25 ▾ 1 - 3 of 3 ↺ ⏪ ⏩ ▾ ☰

_id: ObjectId('6a8d08519998e4beab722ed0')

title: "Quantum Horizons"

author: "R. Menon"

genre: "Science"

publishedYear: 2023

formats: Array (2)

specs: Object (4)

avgRating: 4.9

reviews: Array (2)

_id: ObjectId('6a8d08239998e4beab722eca')

title: "The Silent Algorithm"

author: "R. Menon"

genre: "Technology"

publishedYear: 2022

formats: Array (3)

specs: Object (5)

avgRating: 4.6

reviews: Array (2)